

SUMMARY TABLE: BIOLOGY GRADE 11

Sub-Strand 2.1: Reproduction in Plants — Teacher Reference (Pre-filled)

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Sub-Strand	Sub-Strand 2.1: Reproduction in Plants
Driving Question	DRIVING QUESTION: Why do some farm plants produce offspring without seeds while others require seeds and a partner tree to make fruit?

INSTRUCTIONS

FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 Cuttings, Tubers and Lonely Trees: Why Do Plants Reproduce So Differently?	Students predicted first, then examined real or pictured samples: cassava stem cuttings, sugarcane setts, sweet potato vines, ginger rhizomes and onion bulbs sprouting, set beside a lone pawpaw tree flowering with no fruit. Teacher note: allow WAIT TIME of 10 seconds after asking 'What do the sprouting parts have in common?' and cold-call 3 students. Expect them to notice that new shoots and roots appear from stems, rhizomes or bulbs with no seed and no flower involved.	Plants such as cassava, sugarcane, sweet potato, ginger and onion reproduce asexually (vegetative propagation) from stems, rhizomes, bulbs or vine cuttings. One parent is enough, and the offspring are genetically identical to it. Other plants, like pawpaw and avocado, rely on flowers, seeds and sometimes a second plant. Teacher note: correct the common idea that every new plant must come from a seed.	Cassava farmers in Western Kenya plant stem cuttings because the cutting contains cells that divide by mitosis to form roots and shoots, so no partner is needed. A lone pawpaw tree may flower heavily but cannot make fruit if it is only one sex or cannot pollinate itself, so it needs a partner. This gives students a first answer to the driving question: the mode of reproduction decides whether a partner is needed. Record on the DQB new questions such as 'How does pollination work?' and start the model with two pathways, asexual (one parent) and sexual (flowers, seeds).
Lesson 2 Flower Detectives: Classifying Flowers by Ovary Position, Symmetry, Inflorescence and Sex	Students dissected or examined flowers available locally (hibiscus, bean or pea, maize tassel and silk, pawpaw male and female flowers, sukuma wiki or kale flowers, Tithonia). They used hand lenses, sketched, and sorted flowers into groups. Teacher note: ask 'Where is the ovary in relation to the petals and sepals?' and wait 8 seconds before taking answers. Circulate and check that students cut longitudinally to see ovary position.	Flowers are classified by ovary position (superior, where the ovary sits above the attachment of the other parts, or inferior, where it sits below), symmetry (actinomorphic, which has many planes of symmetry, or zygomorphic, which has one), inflorescence (single flower or a cluster), and sex (bisexual, with both stamens and carpels, or unisexual, with either). Pawpaw can have separate male and female flowers, and often separate male and	Classification shows that flower structure is linked to how pollination happens. Unisexual flowers on separate trees, like many pawpaws, make a partner tree essential, while bisexual flowers allow other options. This explains why a lone pawpaw may flower but not fruit if it is male, or if it is female with no pollen source. Add to the DQB: 'Which flower features stop self-pollination?' Update the model by labelling male and female parts and adding the sex

		female trees. Teacher note: stress that bisexual does not mean self-pollinating.	of the flower as a factor that determines fruit set.
Lesson 3 Why Can a Lone Pawpaw Tree Flower Heavily but Never Set Fruit? Adaptations That Prevent Self-Pollination	Students studied diagrams, flower samples and case data on pawpaw, avocado, maize and other plants to identify adaptations. They compared flowers where stamens and stigma mature at different times, where they sit at different heights, where the sexes are on separate plants, and where pollen is chemically rejected. Teacher note: use a Think-Pair-Share and cold-call 4 pairs for the adaptation they found.	Four types of adaptations prevent self-pollination and promote cross-pollination: timing (dichogamy, where anthers and stigma mature at different times; avocado flowers open as female then male on a daily cycle), structural (different heights of stamens and stigma, as in heterostyly), separate sexes (dioecious plants like many pawpaws), and chemical (self-incompatibility, where pollen is rejected by the stigma or style). Teacher note: emphasise that these promote variation.	A lone pawpaw that is male has no ovules and cannot form fruit. A lone female has ovules but no pollen. A lone avocado may flower but its male and female phases are out of step, so it fruits poorly without a second tree of the complementary type. Nairobi-area and Murang'a farmers plant mixed avocado types for this reason. So flowering without fruiting is explained by prevented self-pollination. Add DQB questions about what happens after pollen lands, and add adaptations to the model.
Lesson 4 Following the Pollen Tube: How Double Fertilisation Explains Why a Lone Pawpaw Flowers but Sets No Fruit	Students used diagrams, a simulation or microscope slides of germinating pollen to follow pollen from the stigma, down the style via the pollen tube, to the ovule's micropyle. They traced the two sperm nuclei and labelled the egg cell and the two polar nuclei. Teacher note: pause the animation and ask 'Where does each sperm nucleus go?' with 10 seconds of WAIT TIME.	In double fertilisation one sperm nucleus fuses with the egg cell to form the zygote (the future embryo), and the other fuses with the two polar nuclei to form the triploid endosperm, the food store. After pollination the pollen grain grows a pollen tube that carries the male nuclei to the ovule. Teacher note: remind students the zygote is diploid and the endosperm is triploid, and this is a feature of flowering plants only.	Without compatible pollen reaching the stigma, no pollen tube grows and no fertilisation occurs, so the ovules do not develop into seeds and the flower drops off. This is why a lone pawpaw tree can flower heavily but set no fruit. In maize, each kernel on a cob needs its own pollen grain and silk, so poor pollination gives gaps on the cob. Add to the DQB the question 'What does the flower become after fertilisation?' and revise the model to include the pollen tube and the two fusions.
Lesson 5 Inside the Fruit: Placentation as Evidence of Post-Fertilisation Development	Students cut open and compared local fruits such as beans or pods, tomato, pawpaw, orange and maize cobs, noting how seeds attach inside the ovary. They sketched cross-sections and matched them to placentation types. Teacher note: provide knives only under supervision; ask 'Where are the seeds attached?' and cold-call 3 students.	After fertilisation the ovule becomes the seed and the ovary becomes the fruit. Placentation is how ovules attach in the ovary: marginal (along one edge, as in bean pods), axile (on a central axis where the ovary is divided into chambers, as in tomato and orange), parietal (on the ovary wall, as in pawpaw), and free-central (on a central column with no partitions). Teacher note: placentation can be checked in the flower before fertilisation and the fruit afterwards.	Seeds and fruit are evidence that fertilisation occurred, because only fertilised ovules develop into seeds and the ovary wall develops into the fruit. A pawpaw with parietal placentation shows many seeds attached to the wall, each from a fertilised ovule, so many pollen grains were needed. A lone pawpaw with no fertilisation will produce no fruit. Record DQB questions on advantages and disadvantages of seeds versus cuttings, and add fruit and seed development to the model.
Lesson 6 Variation or Copies? Debating the Trade-offs of Sexual and Asexual Reproduction in Kenyan Crops	Students took roles as farmers, plant breeders and agricultural officers in a structured debate on crops such as cassava, sugarcane, bananas, maize, beans and avocado. They used evidence cards on yield, disease outbreaks and cost.	Asexual reproduction needs only one parent and gives fast, identical offspring that are useful for colonising an area and keeping good traits, but a lack of variation means all plants can be wiped out by the same disease or pest, as with cassava	Both methods are useful depending on the goal. A farmer in Kisumu uses cassava cuttings for quick, uniform harvests, but a plant breeder uses seeds from crosses to produce new disease-resistant varieties. Pawpaw and avocado rely on sexual

	Teacher note: give 5 minutes for preparation, require every student to cite one piece of evidence, and use a 'Claim-Evidence-Reasoning' sheet.	mosaic disease. Sexual reproduction needs two gametes and often two parents, is slower, and produces variation that helps adaptation to changing conditions and disease resistance, and allows seed dispersal. Teacher note: avoid labelling either as 'better'.	reproduction, so a partner tree is needed to produce viable seeds and fruit. This links back to the driving question by showing why plants have evolved different strategies. Update the DQB and the model with a trade-off table.
Lesson 7 Putting the Evidence Together: Why Cassava Cuttings Need No Seeds but Lone Pawpaw and Avocado Trees Need a Partner	Students reviewed all their evidence from the earlier lessons, including cuttings, flower dissections, adaptation cases, pollen tube diagrams and fruit sections, and used it to build a claim-evidence-reasoning explanation. Teacher note: set out stations with each piece of evidence, give 12 minutes, and cold-call 4 groups to share while others check for gaps.	Asexual reproduction by cuttings and other vegetative parts gives identical plants quickly without flowers or pollination. Sexual reproduction needs flowers, pollination, double fertilisation and fruit and seed formation, and many plants have adaptations that prevent self-pollination, so a partner is needed. Teacher note: check that students link each step to an observation, not just recite facts.	A cassava cutting grows by mitosis from the parent's cells, so it needs no seeds or partner. A lone pawpaw or avocado needs compatible pollen from another tree because of separate sexes, timing differences or chemical incompatibility, and without pollination and fertilisation no seeds or fruit form. Students should be able to explain the whole chain from flower to fruit. Use the DQB to mark which questions are now answered and which remain, and revise the model so that both pathways are complete.
Lesson 8 Final Explanation: Why Cassava Cuttings Grow Without Seeds but a Lone Pawpaw Needs a Partner Tree	Students completed their final model and written explanation, then presented to peers and compared it with their first predictions from Lesson 1. They used peer feedback checklists. Teacher note: give 10 seconds of WAIT TIME after each presentation, and ask the audience to name one strength and one improvement.	Asexual reproduction (vegetative propagation) forms identical offspring (clones) by mitosis, is fast and needs no partner, but offers little variation. Sexual reproduction involves flowers, pollination, double fertilisation and the development of seeds and fruit, and gives variation, with adaptations that promote cross-pollination. Teacher note: look for correct use of terms such as zygote, endosperm, ovary and pollen tube.	Cassava, sugarcane and sweet potato can be grown from cuttings because plant cells can divide and form new roots and shoots, so one parent is enough. A lone pawpaw or avocado flowers but gives no fruit because self-pollination is prevented by separate sexes, timing or chemistry, and without pollen from another tree there is no pollen tube, no double fertilisation, and no seeds or fruit. This fully answers the driving question. Encourage students to apply the idea to a new crop, such as mango or banana, as an extension.

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